

BRIEF NOTES FOR MTH6130 PROBABILITY III

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- What is in these notes: definitions, theorems, proofs, methods.
- What is not in these notes: examples, motivation, chit-chat.

1. DISCRETE MARKOV CHAINS

Definition 1.1. A *stochastic process* is a collection of random variables (r.v.'s) $\{X_t : t \in T\}$ indexed by a set T (usually thought of as “time”). The X_t take values in some *state space* S .

In the first half of the course we take $T = \mathbb{N}$, so the stochastic process is $X_0, X_1, X_2, \dots$. This is a *discrete-time* process. Later we consider continuous time $T = \mathbb{R}_{\geq 0} = \{x \in \mathbb{R} : x \geq 0\}$.

1.1. The Markov property.

Definition 1.2. A stochastic process $X_0, X_1, X_2, \dots$ on a (finite or countably infinite) state space S is a *Markov chain* if it satisfies, for all $t \in \mathbb{N}$,

$$\mathbb{P}(X_{t+1} = s \mid X_t = s_t, X_{t-1} = s_{t-1}, \dots, X_0 = s_0) = \mathbb{P}(X_{t+1} = s \mid X_t = s_t),$$

for all $s, s_t, s_{t-1}, \dots, s_0 \in S$ for which the conditional probabilities are defined.

This is called the *Markov property*. Informally: “The future conditioned on the present does not depend on the past.”

If $\mathbb{P}(X_{t+1} = j \mid X_t = i)$ is independent of t then we say that the Markov chain is *homogeneous*. (We only consider homogeneous Markov chains in this course.) Then we write $\mathbb{P}(X_{t+1} = j \mid X_t = i) = p_{ij}$ and call p_{ij} the *transition probabilities*.

Lemma 1.1. If $X_0, X_1, X_2, \dots$ is a Markov chain with $\mathbb{P}(X_0 = i) = \mu_i$ for $i \in S$, then

$$\mathbb{P}(X_0 = s_0, X_1 = s_1, X_2 = s_2, \dots, X_t = s_t) = \mu_{s_0} p_{s_0 s_1} p_{s_1 s_2} \cdots p_{s_{t-1} s_t},$$

for all $s_0, s_1, \dots, s_t \in S$.

These notes are condensed and adapted from lecture notes of Dr Robert Johnson.

Proof.

$$\begin{aligned}
& \mathbb{P}(X_0 = s_0, X_1 = s_1, \dots, X_t = s_t) \\
&= \mathbb{P}(X_0 = s_0) \times \mathbb{P}(X_1 = s_1 \mid X_0 = s_0) \times \mathbb{P}(X_2 = s_2 \mid X_1 = s_1, X_0 = s_0) \\
&\quad \times \cdots \times \mathbb{P}(X_t = s_t \mid X_{t-1} = s_{t-1}, \dots, X_0 = s_0) \\
&= \mu_{s_0} \mathbb{P}(X_1 = s_1 \mid X_0 = s_0) \mathbb{P}(X_2 = s_2 \mid X_1 = s_1) \cdots \mathbb{P}(X_t = s_t \mid X_{t-1} = s_{t-1}) \\
&\quad \text{(by the Markov property)} \\
&= \mu_{s_0} p_{s_0 s_1} p_{s_1 s_2} \cdots p_{s_{t-1} s_t} \\
&\quad \text{(by definition of } p_{ij} \text{).}
\end{aligned}$$

□

A convenient means to visualise Markov chains is by *transition graphs*. The vertex set of a transition graph of a Markov chain is the set of states S . For each pair of states i, j for which the transition probability p_{ij} is non-zero, there is a directed edge (arc) from vertex i to vertex j labelled p_{ij} .

1.2. Transition matrices. The events $X_t = s$, as s ranges over S , partition the sample space. Thus

$$\sum_{j \in S} \mathbb{P}(X_t = j) = 1$$

and

$$\sum_{j \in S} \mathbb{P}(X_t = j \mid X_{t-1} = i) = 1,$$

for all $i \in S$, i.e.,

$$\sum_{j \in S} p_{ij} = 1.$$

The *transition matrix* is the matrix P with rows and columns indexed by S and with ij th entry p_{ij} . We have just seen that it is *stochastic*, i.e., that all row sums are 1. For convenience in numbering the rows and columns of the matrix P , we usually take $S = \{1, 2, \dots, n\}$.

The r -step transition probabilities are

$$p_{ij}^{(r)} = \mathbb{P}(X_{t+r} = j \mid X_t = i).$$

(These are independent of t as the chain is homogeneous.) Note that

$$p_{ij}^{(0)} = \begin{cases} 1, & \text{if } i = j; \\ 0 & \text{otherwise} \end{cases} \quad \text{and} \quad p_{ij}^{(1)} = p_{ij}.$$

Theorem 1.2 (Chapman-Kolmogorov relations).

$$p_{ij}^{(r+s)} = \sum_{k \in S} p_{ik}^{(r)} p_{kj}^{(s)}, \quad \text{for all } i, j \in S \text{ and } r, s \geq 0.$$

Proof.

$$\begin{aligned}
p_{ij}^{(r+s)} &= \mathbb{P}(X_{r+s} = j \mid X_0 = i) \\
&= \sum_{k \in S} \mathbb{P}(X_{r+s} = j, X_r = k \mid X_0 = i) \\
&\quad \text{(since the events } X_r = k \text{ partition the sample space)} \\
&= \sum_{k \in S} \mathbb{P}(X_r = k \mid X_0 = i) \mathbb{P}(X_{r+s} = j \mid X_r = k, X_0 = i) \\
&= \sum_{k \in S} \mathbb{P}(X_r = k \mid X_0 = i) \mathbb{P}(X_{r+s} = j \mid X_r = k) \\
&\quad \text{(by the Markov property)} \\
&= \sum_{k \in S} p_{ik}^{(r)} p_{ik}^{(s)}.
\end{aligned}$$

□

Corollary 1.3. *If P is the transition matrix of a Markov chain, then P^r is the matrix of r -step transition probabilities, i.e.,*

$$P^r = \begin{pmatrix} p_{11}^{(r)} & p_{12}^{(r)} & \cdots & p_{1n}^{(r)} \\ p_{21}^{(r)} & p_{22}^{(r)} & \cdots & p_{2n}^{(r)} \\ \vdots & \vdots & & \vdots \\ p_{n1}^{(r)} & p_{n2}^{(r)} & \cdots & p_{nn}^{(r)} \end{pmatrix}.$$

Proof. The proof is by induction on r , with the base case $r = 1$ being immediate. If $r > 1$ then

$$\begin{aligned}
(P^r)_{ij} &= (P^{r-1}P)_{ij} \\
&= \sum_{k \in S} (P^{r-1})_{ik} P_{kj} \\
&= \sum_{k \in S} p_{ik}^{(r-1)} p_{kj}^{(1)} && \text{(by inductive hypothesis)} \\
&= p_{ij}^{(r)} && \text{(by Theorem 1.2).}
\end{aligned}$$

□

If the matrix P is diagonalisable, i.e., there is an invertible matrix M and a diagonal matrix D such that $P = MDM^{-1}$, then

$$P^r = \underbrace{(MDM^{-1})(MDM^{-1}) \cdots (MDM^{-1})}_{r \text{ copies}} = MD^r M^{-1},$$

which provides an easy way to compute P^r .

1.3. First-step analysis. A state is *absorbing* if $p_{ii} = 1$; when we reach an absorbing state we never leave it.

Several quantities — what is the probability of reaching a certain absorbing state? what is the expected time to absorption? — can be computed by conditioning on the first step. The computation reduces to solving simultaneous linear equations.

Theorem 1.4. *Suppose (X_t) is a Markov chain with state space S and absorbing states $A \subseteq S$, and suppose $w : S \setminus A \rightarrow \mathbb{R}$ is a weight function on non-absorbing states. Let $T = \min\{t : X_t \in A\}$ and $W = \sum_{t=0}^{T-1} w(X_t)$.*

(1) *Fix $k \in A$. Let $a_i = \mathbb{P}(X_T = k \mid X_0 = i)$. Then the a_i satisfy*

$$a_i = \begin{cases} 1, & \text{if } i = k; \\ 0, & \text{if } i \in A \setminus \{k\}; \\ p_{ik} + \sum_{j \in S \setminus A} p_{ij} a_j & \text{if } i \in S \setminus A. \end{cases}$$

(2) *Let $w_i = \mathbb{E}(W \mid X_0 = i)$. Then the w_i satisfy*

$$w_i = \begin{cases} 0, & \text{if } i \in A; \\ w(i) + \sum_{j \in S \setminus A} p_{ij} w_j & \text{if } i \notin A. \end{cases}$$

(3) *If S is finite and from every state it is possible to reach an absorbing state then the above systems of equations have unique solutions.*

Proof. (1) If $i \in A$ (i.e., we start in an absorbing state) then $T = 0$; thus $a_k = \mathbb{P}(X_T = k \mid X_0 = k) = 1$, and if $i \in A \setminus \{k\}$ then $a_i = \mathbb{P}(X_T = k \mid X_0 = i) = 0$. Otherwise (i.e., if $i \notin A$), conditioning on the first step:

$$\begin{aligned} a_i &= \sum_{j \in S} \mathbb{P}(X_1 = j \mid X_0 = i) \mathbb{P}(X_T = k \mid X_1 = j) \\ &\quad \text{(Law of Total Probability)} \\ &= \sum_{j \in S} p_{ij} a_j \\ &= p_{ik} + \sum_{j \in S \setminus A} p_{ij} a_j \end{aligned}$$

(2) If $i \in A$ then $T = 0$ and $w_i = 0$. Otherwise, conditioning on the first step:

$$\begin{aligned} w_i &= w(i) + \sum_{j \in S} \mathbb{P}(X_1 = j \mid X_0 = i) \mathbb{E}\left(\sum_{t=1}^{T-1} w(X_t) \mid X_1 = j\right) \\ &\quad \text{(Law of Total Expectation)} \\ &= w(i) + \sum_{j \in S} p_{ij} w_j \\ &= w(i) + \sum_{j \in S \setminus A} p_{ij} w_j. \end{aligned}$$

(3) Outside the scope of the module. □

1.4. Long-term behaviour. A *probability vector* is a row vector $\boldsymbol{\mu} = (\mu_1, \mu_2, \dots, \mu_n)$ with $\mu_i \geq 0$ and $\sum_{i=1}^n \mu_i = 1$. Let $X_0, X_1, \dots$ be a Markov chain with $S = \{1, 2, \dots, n\}$. Let the distribution of X_0 be given by a probability vector $\boldsymbol{\mu}^{(0)} = (\mu_1^{(0)}, \mu_2^{(0)}, \dots, \mu_n^{(0)})$ (i.e., $\mathbb{P}(X_0 = k) = \mu_k^{(0)}$), the *initial distribution*. Let $\boldsymbol{\mu}^{(t)}$ be the distribution of X_t (i.e., $\mathbb{P}(X_t = k) = \mu_k^{(t)}$).

Lemma 1.5. *With the above definitions, $\boldsymbol{\mu}^{(t)} = \boldsymbol{\mu}^{(0)} P^t$.*

Proof.

$$\mu_k^{(t)} = \mathbb{P}(X_t = k) = \sum_{i=1}^n \mathbb{P}(X_t = k \mid X_0 = i) \mathbb{P}(X_0 = i) = \sum_{i=1}^n \mu_i^{(0)} p_{ik}^{(t)}.$$

Now appeal to Corollary 1.3. □

The general two-state chain. Suppose $S = \{1, 2\}$ and

$$P = \begin{pmatrix} 1 - \alpha & \alpha \\ \beta & 1 - \beta \end{pmatrix}.$$

Boring cases are

- $\alpha = \beta = 0$, when $P = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, $P^t = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, and $\lim_{t \rightarrow \infty} P^t = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$; and
- $\alpha = \beta = 1$, when $P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $P^t = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ if t is even, and $P^t = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ if t is odd. In this case $\lim_{n \rightarrow \infty} P^t$ does not exist.

The interesting case is when $0 < \alpha + \beta < 2$. Then P has eigenvalue 1 with eigenvector $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$, and $\lambda = 1 - \alpha - \beta$ with eigenvector $\begin{pmatrix} \alpha \\ -\beta \end{pmatrix}$. Thus

$$P = \begin{pmatrix} 1 & \alpha \\ 1 & -\beta \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & \lambda \end{pmatrix} \begin{pmatrix} \frac{\beta}{\alpha + \beta} & \frac{\alpha}{\alpha + \beta} \\ \frac{1}{\alpha + \beta} & \frac{-1}{\alpha + \beta} \end{pmatrix},$$

and so

$$\begin{aligned} P^t &= \frac{1}{\alpha + \beta} \begin{pmatrix} 1 & \alpha \\ 1 & -\beta \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & \lambda^t \end{pmatrix} \begin{pmatrix} \beta & \alpha \\ 1 & -1 \end{pmatrix} \\ &= \frac{1}{\alpha + \beta} \begin{pmatrix} 1 & \alpha \\ 1 & -\beta \end{pmatrix} \begin{pmatrix} \beta & \alpha \\ \lambda^t & -\lambda^t \end{pmatrix} \\ &= \frac{1}{\alpha + \beta} \begin{pmatrix} \beta & \alpha \\ \beta & \alpha \end{pmatrix} + \frac{\lambda^t}{\alpha + \beta} \begin{pmatrix} \alpha & -\alpha \\ -\beta & \beta \end{pmatrix}. \end{aligned}$$

Since $-1 < \lambda < 1$,

$$\lim_{t \rightarrow \infty} P^t = \frac{1}{\alpha + \beta} \begin{pmatrix} \beta & \alpha \\ \beta & \alpha \end{pmatrix} = W,$$

say, and

$$\lim_{t \rightarrow \infty} \boldsymbol{\mu}^{(t)} = \lim_{t \rightarrow \infty} \boldsymbol{\mu}^{(0)} P^t = \boldsymbol{\mu}^{(0)} W = \left(\frac{\beta}{\alpha + \beta}, \frac{\alpha}{\alpha + \beta} \right),$$

for any initial distribution $\boldsymbol{\mu}^{(0)}$. Note that, in the limit as $t \rightarrow \infty$, the distribution of X_t does not depend on the initial distribution of X_0 : this is because $\lim_{t \rightarrow \infty} P^t$ exists (which is not so when $\alpha = \beta = 1$) and has equal rows (which is not so when $\alpha = \beta = 0$).

Definition 1.3. A probability vector $\boldsymbol{w} = (w_1, \dots, w_n)$ is a *limiting distribution* for P if

$$P^t \rightarrow \begin{pmatrix} w_1 & w_2 & \dots & w_n \\ w_1 & w_2 & \dots & w_n \\ \vdots & & \ddots & \\ w_1 & w_2 & \dots & w_n \end{pmatrix}, \quad \text{as } t \rightarrow \infty.$$

It is an *equilibrium distribution* if $\boldsymbol{w}P = \boldsymbol{w}$.

For the 2-state example,

- If $\alpha = \beta = 0$ then any vector $(w, 1 - w)$ is an equilibrium distribution, and there is no limiting distribution.
- If $\alpha = \beta = 1$ then $(\frac{1}{2}, \frac{1}{2})$ is the unique equilibrium distribution, but there is no limiting distribution.
- If $0 < \alpha + \beta < 2$ then $(\frac{\beta}{\alpha + \beta}, \frac{\alpha}{\alpha + \beta})$ is the unique equilibrium distribution and the limiting distribution.

Definition 1.4. A Markov chain is *irreducible* if, for all $i, j \in S$, there is a $k \geq 0$ with $p_{ij}^{(k)} > 0$; it is *regular* if there is a $k \geq 0$ with $p_{ij}^{(k)} > 0$ for all $i, j \in S$.

Theorem 1.6. Let P be the transition matrix of a regular Markov chain on a finite state space. Then P has a limiting distribution $\boldsymbol{w}$. Moreover, $\boldsymbol{w}$ is the unique equilibrium distribution for P .

Proof. We rely on the Perron-Frobenius Theorem, whose proof is beyond the scope of the course.

Since the Markov chain is regular, we know that there exists $r \in \mathbb{N}$ such that $p_{ij}^{(r)} > 0$ for all $i, j \in S$. By Corollary 1.3, $(P^r)_{ij} = p_{ij}^{(r)} > 0$, for all $i, j \in S$. A matrix that has this property — namely that there exists r such that $(P^r)_{ij} > 0$, for all $i, j \in S$ — is said to be *regular*. (The transition matrix of a regular Markov chain is regular, but not every regular matrix is the transition matrix of an regular Markov chain.)

The Perron-Frobenius Theorem for non-negative, regular matrices states (among other things) that there exists a unique (up to scaling) row vector $\boldsymbol{v}$ satisfying $P\boldsymbol{v}^T = r\boldsymbol{v}^T$ (i.e., $\boldsymbol{v}^T$ is a right eigenvector of P), and a unique positive row vector $\boldsymbol{w}$ satisfying $\boldsymbol{w}P = r\boldsymbol{w}$ (i.e., $\boldsymbol{w}$ is a left eigenvector of P). (A vector is said to be positive if all its entries are positive.) Note that the eigenvalue r is the same in both cases. Now it is easy to check that

$$P \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix},$$

so in our case $\boldsymbol{v} = (1, 1, \dots, 1)$ and $r = 1$. So we deduce that there is a unique positive row vector $\boldsymbol{w}$ such that $\boldsymbol{w}P = r\boldsymbol{w} = \boldsymbol{w}$. Scale $\boldsymbol{w}$ so that it is a probability vector, i.e., so that its entries sum to 1. Then $\boldsymbol{w}$ is the unique equilibrium distribution of P .

Another consequence of the Perron-Frobenius Theorem is that

$$\lim_{t \rightarrow \infty} P^t = r \mathbf{v}^T \mathbf{w} = \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix} (w_1, w_2, \dots, w_n) = \begin{pmatrix} w_1 & w_2 & \dots & w_n \\ w_1 & w_2 & \dots & w_n \\ \vdots & & \ddots & \\ w_1 & w_2 & \dots & w_n \end{pmatrix},$$

assuming (as here) that $\mathbf{w}$ and $\mathbf{v}$ are normalised so that $\mathbf{w} \mathbf{v}^T = 1$. Thus, $\mathbf{w}$ is also the limiting distribution of P . $\square$

Fix $k \in S$ and define the indicator r.v.

$$A_s = \begin{cases} 1, & \text{if } X_s = k; \\ 0, & \text{otherwise.} \end{cases}$$

Then $\mathbb{E}(A_s) = \mathbb{P}(X_s = k)$. The proportion of time spent in state k up to but not including time t is $\frac{1}{t} \sum_{s=0}^{t-1} A_s$. We are interested in the limit $\lim_{t \rightarrow \infty} \mathbb{E} \left(\frac{1}{t} \sum_{s=0}^{t-1} A_s \right)$, assuming it exists: this is the expected proportion of time spent in state k in the long term.

Lemma 1.7. *If $\mathbf{w}$ is the limiting distribution of P (see Definition 1.3) then*

$$\mathbb{E} \left(\frac{1}{t} \sum_{s=0}^{t-1} A_s \right) \rightarrow w_k, \quad \text{as } t \rightarrow \infty.$$

Proof.

$$\mathbb{E} \left(\frac{1}{t} \sum_{s=0}^{t-1} A_s \right) = \frac{1}{t} \mathbb{E} \left(\sum_{s=0}^{t-1} A_s \right) = \frac{1}{t} \sum_{s=0}^{t-1} \mathbb{E}(A_s) = \frac{1}{t} \sum_{s=0}^{t-1} \mathbb{P}(X_s = k).$$

But $\mathbb{P}(X_s = k) \rightarrow w_k$ as $s \rightarrow \infty$, so

$$\frac{1}{t} \sum_{s=0}^{t-1} \mathbb{P}(X_s = k) \rightarrow w_k, \quad \text{as } t \rightarrow \infty.$$

(If you took MTH5124 *Convergence and Continuity*, you could pause to verify this claim.) $\square$

Theorem 1.8. *Any irreducible Markov chain on a finite state space has a unique equilibrium distribution $\mathbf{w}$. Moreover,*

$$\mathbb{E}(\text{proportion of time spent in state } k \text{ up to time } t) \rightarrow w_k, \quad t \rightarrow \infty.$$

Proof. Since the Markov chain is irreducible, we know that, for every $i, j \in S$ there exists $r \in \mathbb{N}$ such that $p_{ij}^{(r)} > 0$. By Corollary 1.3, $(P^r)_{ij} = p_{ij}^{(r)} > 0$. A matrix that has this property — namely that for all i, j there exists r such that $(P^r)_{ij} > 0$ — is said to be *irreducible*. (The transition matrix of an irreducible Markov chain is irreducible, but not every irreducible matrix is the transition matrix of an irreducible Markov chain.)

Applied to non-negative, irreducible matrices, The Perron-Frobenius Theorem still assures us that P has a unique positive right eigenvector $\mathbf{v}$, and a unique positive left eigenvector $\mathbf{w}$. As in the proof of Theorem 1.6, it follows that $\mathbf{w}$ is the unique equilibrium

distribution of P . (The second part of the proof of Theorem 1.6 fails for irreducible matrices P , so we cannot deduce that $\mathbf{w}$ is a limiting distribution.)

The claim about the proportion of time spent in state k in the long term is beyond the scope of the module. $\square$

1.5. Recurrence and Transience. The definitions and theorems in the previous section were phrased for finite state spaces. We often want to consider (countably) infinite state spaces. If $S = \{0, 1, 2, \dots\} = \mathbb{N}$ we say that $(w_0, w_1, w_2, \dots)$ is an *equilibrium distribution* if $w_k = \sum_{i=0}^{\infty} w_i p_{ik}$ for all k , and $\sum_{i=0}^{\infty} w_i = 1$. We say that $(w_0, w_1, w_2, \dots)$ is a *limiting distribution* if $\mathbb{P}(X_t = k \mid X_0 = i) \rightarrow w_k$ ($t \rightarrow \infty$), for all $i, k \in \mathbb{N}$.

Definition 1.5. Suppose $a, b \in S$ are states. We say that a *communicates with* b (written $a \rightarrow b$) if $p_{ab}^{(k)} > 0$ for some $k \geq 0$. We say that a and b *intercommunicate* (written $a \leftrightarrow b$) if $a \rightarrow b$ and $b \rightarrow a$. The *communicating classes* of the chain are the equivalence classes of S under $\leftrightarrow$.

Note (i) $a \leftrightarrow a$ for all $a \in S$, (ii) a Markov chain is irreducible if $a \leftrightarrow b$ for all $a, b \in S$. Define

$$f_{ii}^{(t)} = \mathbb{P}(X_t = i, X_{t-1} \neq i, X_{t-2} \neq i, \dots, X_1 \neq i \mid X_0 = i)$$

(informally, $f_{ii}^{(t)}$ is the probability that the first return to i occurs at time t). Then define $f_{ii} = \sum_{t=1}^{\infty} f_{ii}^{(t)}$ (informally, the probability of ever returning to i).

Definition 1.6. We say that a state i is *recurrent* (or *persistent*) if $f_{ii} = 1$ and *transient* otherwise (i.e., if $f_{ii} < 1$).

Lemma 1.9. For all $t \geq 1$, $p_{ii}^{(t)} = \sum_{s=1}^t f_{ii}^{(s)} p_{ii}^{(t-s)}$.

Proof. Condition on the time s of first return to state i :

$$\begin{aligned} p_{ii}^{(t)} &= \sum_{s=1}^t \mathbb{P}(X_1 \neq i, \dots, X_{s-1} \neq i, X_s = i \mid X_0 = i) \mathbb{P}(X_t = i \mid X_s = i) \\ &= \sum_{s=1}^t f_{ii}^{(s)} p_{ii}^{(t-s)}. \end{aligned}$$

$\square$

Thus, if we know the values $p_{ii}^{(t)}$, for $t = 1, 2, \dots$, we can calculate the first return probabilities $p_{ii}^{(t)}$ as follows:

$$\begin{aligned} f_{ii}^{(1)} &= p_{ii}^{(1)} \\ f_{ii}^{(2)} + p_{ii}^{(1)} f_{ii}^{(1)} &= p_{ii}^{(2)}, \quad \text{i.e., } f_{ii}^{(2)} = p_{ii}^{(2)} - p_{ii}^{(1)} f_{ii}^{(1)} \\ f_{ii}^{(3)} + p_{ii}^{(1)} f_{ii}^{(2)} + p_{ii}^{(2)} f_{ii}^{(1)} &= p_{ii}^{(3)}, \quad \text{i.e., } f_{ii}^{(3)} = p_{ii}^{(3)} - p_{ii}^{(2)} f_{ii}^{(1)} - p_{ii}^{(1)} f_{ii}^{(2)}, \end{aligned}$$

etc. Let M_i be the number of returns to i , given that $X_0 = i$, and write $M_i = \infty$ if the Markov chain returns infinitely often.

Lemma 1.10.

$$\mathbb{P}(M_i = \infty) = \begin{cases} 1, & \text{if } i \text{ is recurrent;} \\ 0, & \text{otherwise, i.e., if } i \text{ is transient.} \end{cases}$$

Proof. Observe that $\mathbb{P}(M_i = k) = f_{ii}^k(1 - f_{ii})$, since f_{ii}^k expresses the probability that we return k times to state i , and $(1 - f_{ii})$ that we never return again. So if $f_{ii} < 1$,

$$\sum_{k=0}^{\infty} \mathbb{P}(M_i = k) = \sum_{k=0}^{\infty} f_{ii}^k(1 - f_{ii}) = (1 - f_{ii}) \sum_{k=0}^{\infty} f_{ii}^k = 1,$$

using the formula for the sum of a geometric progression. On the other hand, if $f_{ii} = 1$ then $\mathbb{P}(M_i = k) = f_{ii}^k(1 - f_{ii}) = 0$ for all $k \in \mathbb{N}$, and hence $\mathbb{P}(M_i = \infty) = 1$. $\square$

Lemma 1.11. (1) *State i is recurrent iff $\sum_{t=1}^{\infty} p_{ii}^{(t)} = \infty$.*

(2) *State i is transient iff $\sum_{t=1}^{\infty} p_{ii}^{(t)}$ is finite.*

Proof. Let

$$A_t = \begin{cases} 1, & \text{if } X_t = i; \\ 0, & \text{otherwise.} \end{cases}$$

(So A_t is the indicator r.v. of the event $X_t = i$.) Then

$$(1) \quad \mathbb{E}(M_i) = \mathbb{E}\left(\sum_{t=1}^{\infty} A_t\right) = \sum_{t=1}^{\infty} \mathbb{E}(A_t) = \sum_{t=1}^{\infty} p_{ii}^{(t)}.$$

Now, $\mathbb{P}(M_i \geq k) = f_{ii}^k$. (The r.h.s. of this equation expresses the fact that I must return k times to state i , and I don't care what happens after that.) If i is recurrent then $f_{ii} = 1$ and hence

$$\mathbb{E}(M_i) = \sum_{k=1}^{\infty} \mathbb{P}(M_i \geq k) = \sum_{k=1}^{\infty} f_{ii}^k = \infty.$$

In this case, by (1), $\sum_{t=1}^{\infty} p_{ii}^{(t)}$ is infinite. On the other hand, if i is transient then $f_{ii}^k < 1$, and hence

$$\mathbb{E}(M_i) = \sum_{k=1}^{\infty} \mathbb{P}(M_i \geq k) = \sum_{k=1}^{\infty} f_{ii}^k = \frac{1}{1 - f_{ii}} - 1 = \frac{f_{ii}}{1 - f_{ii}},$$

using the formula for the sum of a geometric progression. In this case, by (1), $\sum_{t=1}^{\infty} p_{ii}^{(t)}$ is finite.

(Note that parts (1) and (2) of the lemma actually say the same thing!) $\square$

Theorem 1.12. *Suppose $a, b \in S$ are states.*

(1) *If $a \leftrightarrow b$ and a is recurrent then b is recurrent.*

(2) *If $a \leftrightarrow b$ and a is transient then b is transient.*

Proof. Since (a) and (b) are equivalent, we just prove (a).

Since $a \rightarrow b$ there exists r with $p_{ab}^{(r)} > 0$; since $b \rightarrow a$ there exists s with $p_{ba}^{(s)} > 0$. Suppose that a is recurrent. Then

$$\sum_{t=1}^{\infty} p_{bb}^{(t)} \geq \sum_{k=1}^{\infty} p_{bb}^{(r+s+k)} \geq \sum_{k=1}^{\infty} p_{ba}^{(s)} p_{aa}^{(k)} p_{ab}^{(r)} = \alpha \sum_{k=1}^{\infty} p_{aa}^{(k)},$$

where $\alpha = p_{ba}^{(s)} p_{ab}^{(r)} > 0$. But $\sum_{k=1}^{\infty} p_{aa}^{(k)} = \infty$ by Lemma 1.11 and the fact that a is recurrent. So $\sum_{k=1}^{\infty} p_{bb}^{(k)} = \infty$ and b is recurrent. $\square$

So we can speak of a communicating class as being recurrent or transient, knowing that all states are the same. We say that recurrence and transience are *class properties*.

If i is recurrent, let $R_i = \min\{t : t \geq 1, X_t = i\}$, where we assume $X_0 = i$. This is the *first return time*. Then

$$\mathbb{P}(R_i = k) = f_{ii}^{(k)}, \text{ and}$$

$$\mathbb{E}(R_i) = \sum_{k=1}^{\infty} k f_{ii}^{(k)}.$$

$\mathbb{E}(R_i)$ is the expected first return time or the expected time between visits to i .

Definition 1.7. Suppose i is a recurrent state. We say that state i is *positive recurrent* if $\mathbb{E}(R_i)$ is finite, and *null recurrent* if $\mathbb{E}(R_i)$ is infinite.

Theorem 1.13. Suppose i is recurrent.

- If $p_{ii}^{(n)} \rightarrow 0$ as $n \rightarrow \infty$ then i is null recurrent.
- If $p_{ii}^{(n)} \not\rightarrow 0$ as $n \rightarrow \infty$ then i is positive recurrent.

Proof. Omitted. $\square$

The above theorem can be used to show that positive/null recurrence are class properties.

Theorem 1.14. An irreducible Markov chain has a unique equilibrium distribution iff it is positive recurrent. In the case that the equilibrium distribution exists, it is given by $w_i = 1/\mathbb{E}(R_i)$ for all $i \in S$.

Proof. Omitted. $\square$